User Stories

Requirements Engineering → User-Centered Requirements → Agile Development

What Is a User Story?

Definition: A User Story is a requirement expressed from the perspective of an end-user goal. It focuses on who needs something, what they need, and why they need it.

- It focuses on the viewpoint of a role who will use or be affected by the solution.
- It expresses the requirement in language meaningful to that role.
- It clarifies the reason for or business value behind the requirement.
- It allows high-level requirements to be captured without prematurely specifying low-level implementation details.
- It encourages collaboration between users, product owners, analysts, developers, and testers.

A User Story is not a complete technical specification. It is a concise placeholder for a conversation about the requirement.

User Story Format:

As a <role>,
I need / I want <goal>,
So that <justification / business value>.

Component	Purpose
As a <role>	Identifies the user or stakeholder.
I need / I want <goal>	States what the user wants to accomplish.
So that <value>	Explains the reason or business value.

User Story Examples:

Project-Level Example

As a Marketing Director, I need to improve customer service **so that we retain our customers.**

Detailed Example

As an Investor, I need to see a summary of my investment accounts **so that I can decide where to focus my attention.**

Library Management System Example

As a student, I want to search for books by title, author, ISBN, or category **so that I can quickly find the book I need.**

User Stories and the 3Cs

The 3Cs are Card, Conversation, and Confirmation.

C	Meaning	Explanation
Card	The User Story	A concise statement of the user's need. Traditionally written on an index card. It is a placeholder, not a complete specification.
Conversation	Discussion	The team and stakeholders discuss details, assumptions, constraints, alternatives, and examples to build shared understanding.
Confirmation	Acceptance Criteria	Defines how the team will know the User Story has been successfully delivered. It becomes the basis for testing and acceptance.

Card: What is the user's need?

Conversation: What details do we need to understand?

Confirmation: How will we know we have delivered it correctly?

Example of the 3Cs

Card — Story ID: STK001 | Story Name: Customer Order

As a Customer, I need to place an order so that I can have food delivered to my house.

Conversation: Questions to Clarify the Story

Question	Decision / Clarification
Can the customer save an order and return it later?	Yes, customers can save an incomplete order.
Can the customer change the order before payment?	Yes, customers can add, remove, or change quantities before payment.
Can the customer see a running total?	Yes, the total updates whenever the order changes.
What happens if an item becomes unavailable?	The system shall notify the customer and remove the unavailable item before payment.
Can the customer cancel the order?	Yes, the customer can cancel before the restaurant accepts the order.

Confirmation — Acceptance Criteria

ID	Category	Acceptance Criterion	How to Verify
AC-01	Functional	The customer shall be able to save an incomplete order and retrieve it later.	Save an incomplete order → log out → log in → retrieve the saved order.
AC-02	Functional	The customer shall be able to add, remove, or change the quantity of items before payment.	Add an item → change quantity → remove item → verify the order is updated.
AC-03	Functional	The system shall automatically update the order total whenever the order contents change.	Add/remove items or change quantity → verify the total changes correctly.
AC-04	Functional	The system shall notify the customer if an ordered item becomes unavailable before payment.	Mark an item unavailable → attempt checkout → verify notification is displayed.
AC-05	Functional	The customer shall be able to cancel the order before the restaurant accepts it.	Place an order → cancel before acceptance → verify the order status changes to "Cancelled."

User Stories vs. Traditional Requirements

Aspect	Traditional Requirement	User Story
Focus	System behavior and constraints	User goal and business value
Perspective	System/business perspective	User/stakeholder perspective
Detail	Can be detailed and formal	Concise; details emerge through conversation
Common context	Plan-driven development / SRS	Agile development / Product Backlog
Testing	Verification against requirements	Acceptance criteria provide confirmation

Epic, Feature, and User Story

Important: A large user need is often called an Epic. An Epic can be decomposed into smaller User Stories that are easier to estimate, implement, and test.

Level	Example	Purpose
Epic	As a student, I want to manage my library activities.	Large user goal; too broad for one iteration.
User Story 1	As a student, I want to search books so that I can find a book quickly.	Small, valuable feature.
User Story 2	As a student, I want to reserve an unavailable book so that I can borrow it later.	Small, valuable feature.

INVEST Model for Well-Constructed User Stories

Bill Wake's INVEST model provides six characteristics of effective User Stories.

Letter	Principle	Meaning
I	Independent	The story should be as independent as possible from other stories so it can be moved or implemented with minimal impact.
N	Negotiable	The story is not a contract; it is a placeholder for discussion and clarification.
V	Valuable	The story should provide clear value to the user or owner and describe a feature rather than a technical task.
E	Estimable	The story should contain enough information for the team to estimate its effort.
S	Small	The story should be small enough to estimate and implement within an appropriate timebox. Large Epics should be split.
T	Testable	The story should be clear and specific enough that acceptance criteria and tests can be created.

Example: Applying INVEST

Poor story: As a student, I want the library system to manage everything so that my library experience is better.

- Too broad — not Small.
- Vague value — not sufficiently Valuable.
- Difficult to estimate — not Estimable.
- Difficult to test — not Testable.

Improved story: As a student, I want to search books by title, author, ISBN, or category so that I can quickly find the book I need.

Quality Checks for User Stories

- The story does not combine, overlap, or conflict with another User Story.
- The story follows applicable organizational and project standards.
- The story can be traced back to business needs and project objectives.
- Related stories for different users are cross-referenced where appropriate.
- The story is clear, concise, valuable, and testable.
- Acceptance criteria are specific enough to confirm successful delivery.

Where Are User Stories Recorded?

In Agile projects, User Stories are commonly recorded in the Product Backlog (or a prioritized requirements list, depending on the project framework). They are prioritized according to business value, urgency, risk, dependencies, and other project considerations.

Classroom Activity

Activity: Convert Functional Requirements into User Stories

Library Management System requirements:

- The system shall allow students to search books.
- The system shall allow students to borrow available books.
- The system shall allow students to reserve unavailable books.
- The system shall allow librarians to generate borrowing reports.

Task: For each requirement, write a User Story using the As a / I want / So that format and provide at least two acceptance criteria.

Summary:

- A User Story expresses a requirement from the perspective of a user goal.
- The standard format is: As a <role>, I need/want <goal>, so that <value>.
- The 3Cs are Card, Conversation, and Confirmation.
- Acceptance criteria provide the Confirmation and become the basis for testing.
- Large Epics should be decomposed into smaller User Stories.
- INVEST helps evaluate whether a User Story is well constructed.
- User Stories emphasize user value and collaboration rather than premature solution details.